Shichao Sun

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Education

Sep. 2017 – Jan. 2020(expected)
Master of Engineering, Fudan University, China
Major in Computer Technology
GPA: 3.43/4.0

Sep. 2013 – Jun. 2017
Bachelor of Management, Fudan University, China
Major in Secrecy Policy and Management
GPA: 3.15/4.0

Research Publications

1. **Sun, S.** & Xie, Z. (2017). Bilstm-based models for metaphor detection. In *National CCF Conference on Natural Language Processing and Chinese Computing (NLPCC 2017)*.

Awards

2018 **Scholarship for Outstanding Students** at Fudan University.
2017 **Outstanding Paper** of CCF NLPCC 2017.
2014 – 2016 **National Scholarship of Encouragement and Support** at Fudan University.

Skills

Language: Native Chinese and fluent English.
Coding: Python, C/C++, LaTeX
Deep Learning Framework: PyTorch

Research Experience

2017 – 2018

Math Word Problems Solving

Input a math word problem(involving in addition, subtraction, multiplication, division, power), output the solution expression.

- Implement the baseline (Seq2Seq model), and obtain better performance than the original paper(EMNLP 2017), as a result of careful data processing, strict rules to avoid generating invaild expression and a better hyperparameter setting.
- Compare performance of different attention mechanisms including Bahdanau attention, Luong attention, Dot attention, Minus attention, Bilinear attention. Design multiway attention network motivated by considering different information from various attention mechanisms.
- Considering the fact that context of the number is more important than its exact value, propose dynamic embedding for numbers at decoding side, and the performance is dramatically improved, approximately 2%.
- Data Augmentation for solution expression by generating equivalent expressions, based on the motivation of making the model learn math knowledge about commutative property and distributive property .
- The code of dynamic embedding and data augmentation is available at: `https://github.com/ShichaoSun/math_pade_da`.
- Propose a novel Seq2Tree model to directly generate an expression tree instead of generating solution expression, originating from the human-like goal-driven method of solving math word problems.
- The paper of Seq2Tree is under review by IJCAI 2019.

2016 – 2017

Metaphor Detection

Classify whether a sentence contains the verb metaphor(like “ignite passion”).

- Model sentence by BiLSTM from word embedding to avoid using hand-crafted feature.
- Design three different BiLSTM-based models motivated by three most common types of metaphor.
- Propose an ensemble model of above three models.
- Source code is available at: `https://github.com/ShichaoSun/metaphor`, and the paper is published in NLPCC 2017.

2015 – 2016

Chinese Word Embedding

Adopting word2vec algorithm to train Chinese Word Embedding and optimize it by constructing a better bag of words from dependency tree.